Appendix 11 — On Gauge Structure (STUB)

Appx_11_On_Gauge_Structure_STUB_vX.X

Status

Trailhead — Required before Standard Model, EFT, and QG integration.

Purpose

To determine whether gauge structure arises as a survivable geometric phenomenon under collapse dynamics, rather than as an imposed algebraic symmetry.

This appendix tests whether gauge connections, curvature, and invariants can be derived from Proofield geometry governed by ζ (constraint), ω (extension), and ξ (selection), or whether they must be assumed axiomatically.

Goal

Establish gauge structure as emergent geometry with survivable invariants, admissible under the Units Firewall and compatible with Atlas of Constants enforcement.

Must Include

1. PF Interpretation of Gauge Concepts

    - What “connection” means in Proofield terms (geometric transport, not group fiat)
    - What “curvature” represents as a collapse-stable geometric response
    - Mapping between PF geometry and conventional gauge language (without importing axioms)

2. Candidate Survivable Invariants

    - Identification of dimensionless quantities potentially admissible as Survivors
    - Criteria for survivorship under recursive collapse (stability, selection, non-tuning)
    - Explicit rejection of unit-bearing or calibrated-only quantities

3. Discriminator Plan

- Clear test separating:
- Geometric-origin gauge structure (emergent, selected, collapse-stable)
- Purely algebraic gauge postulates (symmetry assumed by construction)
- Conditions under which algebraic gauge theory fails PF admissibility

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Outputs

- A Gauge Dictionary translating:
- PF geometry → standard gauge theory terms
- Survivable invariants → admissible physical quantities
- Dependency-ready definitions consumable by:
- Standard Model mapping (Appx_15)
- Renormalization / EFT analysis (Appx_17)
- Quantum–Gravity interface (Appx_18–22)

Primary Falsifier

If gauge structure cannot be derived without importing:

- pre-declared symmetry groups,
- fixed algebraic connections,
- or unit-bearing couplings,

then gauge theory is not emergent under PF, and must be treated as a calibrated overlay rather than a fundamental survivor.

Collapse Consequence

If gauge structure is not emergent from geometry, force unification collapses into symmetry bookkeeping, and downstream claims about SM coherence, running couplings, and QG compatibility lose enforcement.

Notes

- This appendix does not unify forces — it defines the admissible substrate on which unification may later occur.

- Failure here does not falsify 7dU wholesale, but blocks all downstream particle and field-theoretic claims.